

Lab 1-3: Requirements Table

Problem Statement #10 – Student Club Event Ticketing & Budget Portal

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Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall route event budget proposals through a multi-stage approval workflow (Faculty Coordinator → Finance Office → Dean).	High	Pass: Status transitions sequentially upon digital sign-off. Fail: Request reaches Dean without Finance Office sign-off.	Ensures financial accountability and prevents unauthorized release of club funds.
FR-002	Functional	The system shall allow a Club Lead to submit a new event proposal containing event title, date, venue, description, and estimated budget.	High	Pass: Proposal is saved with status “Submitted” only when all mandatory fields are complete and the budget is a positive numeric value. Fail: Proposal is saved with a missing mandatory field or a non-numeric budget.	A complete proposal is essential input for review by the Faculty Coordinator, Finance Office and Dean.
FR-003	Functional	The system shall automatically generate a unique, non-forgeable QR-code ticket for each attendee registration once the related event proposal is fully approved.	High	Pass: Every approved registration produces exactly one QR code with a ticket ID traceable to the event and attendee. Fail: A duplicate QR code is issued, or a ticket is generated for an unapproved event.	Enables secure, scalable entry management without manual guest-list checking.
FR-004	Functional	The system shall allow a Campus Admin to scan a QR-code ticket at the venue check-in point, mark the registration as “Checked-in”, and reject tickets that have already been used.	High	Pass: A valid, unused ticket is scanned once and marked checked-in; a second scan of the same ticket is rejected with a “Duplicate entry” alert. Fail: A used ticket is accepted a second time.	Prevents ticket fraud and duplicate entry at high-footfall events.
FR-005	Functional	The system shall provide Club Lead and Campus Admin users with a real-time dashboard showing the status of each budget proposal (Submitted / Under Review / Approved / Rejected) and remaining budget allocation.	Medium	Pass: Dashboard status updates within 5 seconds of any approver action. Fail: Dashboard shows a stale status more than 5 seconds after an update.	Improves transparency and reduces manual follow-up queries to the admin office.
NFR-001	Nonfunctional (Performance & Security)	The QR-code ticket scanner API shall validate entry authenticity in under 100 ms at venue check-in points.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: 95th-percentile validation latency exceeds 100 ms.	Slow validation causes queues and bottlenecks at gate during peak entry periods.
NFR-002	Nonfunctional (Performance & Scalability)	The system shall support at least 500 concurrent ticket registrations for a single event without transaction response time exceeding 2 seconds.	Medium	Pass: Load testing with 500 concurrent virtual users shows a 95th-percentile response time ≤ 2 s with zero dropped requests. Fail: Response time exceeds 2 s, or requests are dropped, under the same load.	Popular club events see sharp registration spikes right after ticket sales open; the portal must stay responsive.

Note: Rows shaded grey (FR-001, NFR-001) are the sample requirements supplied in the problem statement. All other rows (FR-002–FR-005, NFR-002) are original requirements authored for this deliverable.